DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #01F: PATHS & SUBPATHS (PART 6 — GRANULAR AUDIT OF GRAND DAO WAVE 1B: `MERCANTILE_DAO`, `INK_LITERATI`, `LORE_ARCHIVIST`, `WANDERING_DAO`, `SONIC_HARMONY`, `VENOM_GU`, `SOUL_PUPPETRY`, `PRIMORDIAL_SHIFTER` & ALL 46 SUBPATHS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Position (`01C–01H` Granular 43-Path / 286-Subpath Audit)**:
  - `01A–01B`: Shared Macro Lifecycle, Path Library, Retraining & Stage Equivalence Math.
  - `01C`: Group 1 — 5 Founding Pillars (`XIAN`..`HEART_SOUL` — 65 Subpaths).
  - `01D`: Group 2 — 7 Specialist Expansion Paths (`MECH_PUPPET`..`STRATEGY_GOVERNANCE` — 67 Subpaths).
  - `01E`: Group 3 — 8 Cosmic/Spacetime/Weather/Dharma/Geomancy Paths (`ASTRO_DIVINATION`..`PRIMORDIAL_CHAOS` — 48 Subpaths).
  - **`01F` (This Document)**: Group 4 — **8 Scholarly, Mercantile, Wandering, Sonic, Venom, Puppetry & Primal Grand Dao Paths (`MERCANTILE_DAO` [6], `INK_LITERATI` [6], `LORE_ARCHIVIST` [4], `WANDERING_DAO` [6], `SONIC_HARMONY` [6], `VENOM_GU` [6], `SOUL_PUPPETRY` [6], `PRIMORDIAL_SHIFTER` [6] — 46 Subpaths total)**.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Strict Zero-Free-Resource Conservation**: `MERCANTILE_DAO` contracts, trade appraisals, and `VENOM_GU` feeding must strictly obey `Debit === Credit` against `W.sect.stones`, `npc.inv.stones`, and `W.world.resourcePool.stones`.

3. **100% Emergent NPC Autonomy**: Merchants must negotiate real trade/escrow contracts, Wanderers must crave expeditions and stagnate if cooped up indoors, Archivists must decode tablets and annotate sutras, and Venom Gu masters must feed their symbiotic Gu worms organically.

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1. [PATH-MERC-01] Path #21: MERCANTILE_DAO (วิถีพาณิชย์มหามรรคาและพันธสัญญา — 6 Subpaths)

- **Subpaths**: GOLDEN_ABACUS_ARBITER, BLOOD_CONTRACT_NOTARY, CARAVAN_SYNDICATE_MAGNATE, TREASURE_APPRAISER_SAGE, DEBT_FORECLOSURE_ENFORCER, EQUITY_SHIELD_CHANCELLOR.
- **Three Critical Code & Simulation Bugs**:
    1. **Hidden `-8.3%` Combat Damage Penalty in `COMBAT.strike` (`js/combat.js`)**:
        - `COMBAT.strike` maps `MERCANTILE_DAO` (and `LORE_ARCHIVIST`) onto the **`ARRAY` proficiency multiplier** (`1 + clamp(ARRAY / 100, 0, 1) * 0.14`) instead of `trading` + `ABACUS`!
        - Because `MERCANTILE_DAO` cultivators practice `trading` and `ABACUS` (`ARRAY === 0`), they get **`1.00×`**, which is **worse than the generic `else` fallback (`1.09×`)**!
        - **Fix**: Scale `MERCANTILE_DAO` in `COMBAT.strike` by `1 + clamp(Math.max(M.skillRead(a, 'ABACUS'), M.skillRead(a, 'trading')) / 100, 0, 1) * 0.16`!
    2. **Self-Referential Dummy `CONTRACT_LEDGER` (`settleContract` Uncalled & Zero Real Counterparties!)**:
        - `offerContract` / `acceptContract` / `breachContract` were only wired into a dummy monthly cron (`monthlyAutonomousTick`) escrowing `W.sect.stones` against nobody, while `settleContract` has **zero callers**!
        - **Authentic Closed-Loop Contract Loop**:
            - **Peer Breakthrough/Expedition Backing**: A wealthy `MERCANTILE_DAO` disciple (`n.inv.stones >= 80`) offers an escrow contract to a sectmate preparing for an Expedition or Breakthrough (`principalStones` transferred from Merchant `n.inv.stones` $\to$ Borrower `n.inv.stones`, exact `Debit === Credit`!). When the expedition succeeds, `settleContract` repays principal + agreed loot-stone share from real expedition earnings (`+CONTRACT_LEDGER` gauge & `+CULT.gain` for the Merchant!); if the borrower defaults/flees,

`breachContract` grants the Merchant a `contractDefault` foreclosure claim!

3. **Zero Connection to `external_market.js` & `SECT.CARAVAN_AUCTION` + Zero Trade Cultivation XP**:
    - Wire `TREASURE_APPRAISER_SAGE` into `EXT.tradeSect` / `EXT.tradePersonal` (`+8%` better appraisal spread, blocks fire-sale loss on fresh crafted gear!) and `CARAVAN_SYNDICATE_MAGNATE` into `SECT.CARAVAN_AUCTION.dispatchTradeCaravan` (`+15%` trade margin, `-20%` bandit ambush chance), awarding real `CULT.gain` whenever a `MERCANTILE_DAO` cultivator completes a profitable trade or settles a contract!

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2. `[PATH-LIT-LORE-02]` Paths #25 & #40: `INK_LITERATI` (6 Subpaths) & `LORE_ARCHIVIST` (4 Subpaths)

2.1 `LORE_ARCHIVIST` (วิถีอาลักษณ์หอคัมภีร์และถอดอักขระ — 4 Subpaths)

- **Subpaths**: `ANCIENT_RUBBING_DECODER`, `SUTRA_COMPILER_SAGE`, `TECHNIQUE_MIRROR_SCHOLAR`, `FORBIDDEN_SCROLL_CUSTODIAN`.
- **Verified Bugs & Disconnected Systems**:
    1. **Wrong Skill Scaling in `COMBAT.strike`**: Change `ARRAY` scaling (`1.00×` penalty!) to `SCRIPTURE` + `BRUSH` (`1 + clamp(Math.max(SCRIPTURE, BRUSH) / 100, 0, 1) * 0.16`).
    2. **`TECHNIQUE_MIRROR_SCHOLAR` (สายกระจกคัดลอกเคล็ดวิชา) Has Zero Mid-Fight Decoding Ramp**:
        - Lore states: *"วิเคราะห์และถอดรหัสโครงสร้างวิชาของศัตรูกลางศึก คัดลอกจุดแข็งมาเสริมทีม · จุดอ่อน: ต้องใช้เวลาอ่านท่วงท่าต้นศึก"*.
        - Wire into `COMBAT` (`onEnemyCast` / `strike`): start Round 1 at `-6%` offense while observing; each time an enemy casts a technique, gain `+15` `LORE_INDEX` (`Readable Technique Analysis`, up to 4 stacks: `+6%` evasion vs that enemy's `arch` + `+7%` team counter-damage, and copy 1 non-unique stat buff at `LORE_INDEX >= 60`)!
    3. **Zero Library & Sutra Governance Hooks**:
        - `ANCIENT_RUBBING_DECODER`: `+30%` tablet/fragment research speed in `knowledge.js` and `-25%` ruin cipher failure in `explore.js`.
        - `SUTRA_COMPILER_SAGE`: Autonomously runs `SECT.SOUL_FORGE.annotateHeritageSutra` (`v2.0.31`) when stationed at `scripture` and reduces sectmates' `retraining.js` progress loss by `-25%`!

    - `FORBIDDEN_SCROLL_CUSTODIAN`: Reduces `tabooStrain` buildup by `-35%` and serves as the ideal `FORBIDDEN_VAULT` archivist!

2.2 `INK_LITERATI` (วิถีอักษรสวรรค์และปราชญ์จารึก — 6 Subpaths)

- **Subpaths**: `IMPERIAL_EDICT_CALLIGRAPHER`, `LANDSCAPE_SCROLL_PAINTER`, `SWORD_POEM_IMMORTAL`, `CLASSICS_LAW_ARBITER`, `VERMILION_CRITIQUE_CENSOR`, `HISTORIANS_DESTINY_QUILL`.
- **Verified Bugs & Lore Gaps**:
    1. **Uncalled `issueImperialEdict` (`M1` Twin-Code Drift)**: Consolidate `cultivation.js`'s `issueImperialEdict` with `IMPERIAL_INK_DECREE` in `combat.js`.
    2. **Missing Righteous Scholarly Qi (浩然正气 – ปราณเที่ยงธรรม) Rule**:
        - Lore explicitly states: **"หากจิตใจขาดความเที่ยงธรรม อักษรสวรรค์จะสลายตัวทันที"**.
        - Wire `integrity` & `karmicLedger` into `INK_LITERATI`: upright scholars (`integrity >= 65`, `sinKarma === 0`) gain `+18%` `LITERATI_INK` authority and bonus `censor` anti-corruption detection (`VERMILION_CRITIQUE_CENSOR` strips 1 enemy buff on crit/edict!), whereas corrupt/embezzling `INK_LITERATI` (`integrity < 35` or `embezzled > 0`) suffer `-25%` edict duration & `LITERATI_INK` dissipation!

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3. `[PATH-WANDER-03]` Path #20: `WANDERING_DAO` (วิถีจาริกพิภพและมรรคาคนจร — 6 Subpaths)

- **Subpaths**: `HORIZON_WIND_WALKER`, `RUIN_PATHFINDER_SAGE`, `CARAVAN_ESCORT_VANGUARD`, `WILDERNESS_BEAST_TRACKER`, `CLOUD_STEP_HERMIT`, `REALM_MAPPER_CHRONICLER`.
- **Contextual & Simulation Contradictions**:
    1. **Stay-at-Home "Wanderers" Never Lose `ROUTE_INSIGHT` & Cushions Beat Travelling!**:
        - Lore states: **"บ่มเพาะตบะผ่านการก้าวเดินสำรวจโลก... ความเสี่ยงสูงสุด: หากหยุดนิ่งอยู่แต่ในสำนักนานเกินไป ญาณทัศนะคนจรจะค่อยๆ จางลง"**.
        - In runtime, `ROUTE_INSIGHT` never decays if the NPC stays inside the sect for decades, and sitting on a shrine cushion (`A.cultivate`) still gives steady XP while `cognition.js` has no wanderlust drive!
        - **Emergent Fix**:
            - Track `n.ext.lastExpeditionDay`; after `25` consecutive days without leaving the sect (`!n.away`) or patrolling the outer wilds, decay `ROUTE_INSIGHT` by `-2/day` and apply `-15%` indoor

cushion `CULT.rate`, while boosting `EX.tryVolunteer` / `MICRO_LOCATIONS` / Clue-Quest (`VC`) utility by **+35 ("ชีพจรลงเท้าอยากออกท่องยุทธภพ")**!
    - Grant a major **Journey Epiphany (แจ้งชัดหมื่นลี้: +18% of stage threshold `CULT.gain`)** whenever a `WANDERING_DAO` cultivator returns alive from a World-Map expedition, Secret Realm, or Clue Quest!
2. **Subpath Field Hooks**:
    - `CARAVAN_ESCORT_VANGUARD`: Prioritized for caravan/convoy escort (`-30%` ambush loss).
    - `WILDERNESS_BEAST_TRACKER`: `+35%` mutated beast organ discovery (`CONTENT.rareOrgans`) & tracking.
    - `REALM_MAPPER_CHRONICLER`: Shares discovered regional hazard intel with the whole sect library upon returning (`Tier-2 Hub Knowledge` in `02A`!), reducing future sect parties' hazard rates in that region by `-12%`!

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4. `[PATH-SONIC-04]` Path #16: `SONIC_HARMONY` (วิถีเสียงประสานและพิณยุทธ — 6 Subpaths)

- **Subpaths**: `SEVEN_STRINGS_ANNIHILATION`, `JADE_FLUTE_SOUL_CHARMER`, `WAR_DRUM_TEMPEST_HERALD`, `RESONANT_BELL_EXORCIST`, `SILENT_WAVE_ASSASSIN`, `CELESTIAL_CHOIR_CONDUCTOR`.
- **Mechanical & Station Defects**:
    1. **Missing 3-Beat Harmonic Crescendo & Acoustic Interrupt Backlash**:
        - Differentiate `SONIC_HARMONY` cleanly from `XIAN.SOUND_ZITHER` and `SAGE_SUPPORT.CELESTIAL_ZITHER_ENVOY` via its `HARMONIC_CHORD` gauge: each sonic skill advances a **3-Beat Measure (`1 -> 2 -> 3 Finale`)**; Beat 3 unleashes an armor-bypassing **Harmonic Resonance Wave** (`+25%` `mdef`-piercing sonic pulse + `+2` tick cooldown disruption via `SONIC_TEMPO_BREAK`)!
        - Enforce `supremeRisk`: if stunned or silenced (`stun`/`silence`) while `HARMONIC_CHORD >= 60`, trigger **Broken String Backlash** (สายพิณขาดสะท้อนโสตประสาท: drops `HARMONIC_CHORD` to `0` and applies minor self-daze).
    2. **Wrong Sect Station Routing (`ai.js`)**: Route `SONIC_HARMONY` cultivation/practice to `harmonic_pavilion` and `tea` (instead of crowding `shrine`!), radiating passive stress & `heartDemon` relief to nearby listeners!
    3. **Subpath Signatures**: `WAR_DRUM_TEMPEST_HERALD` grants `FIST`/`STAFF`/drum shockwaves + party `fear` immunity; `SILENT_WAVE_ASSASSIN` grants

single-target high-crit NEEDLE/ZITHER stealth assassination without waking/aggroing nearby bystanders!

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5. [PATH-VENOM-05] Path #17: VENOM_GU (วิถีพิษและหนอนกู่โบราณ — 6 Subpaths)

- **Subpaths**: GOLDEN_SILKWORM_GU_LORD, MYRIAD_VENOM_APOTHECARY, BLOOD_PARASITE_HEXER, CORROSIVE_MIASMA_WEAVER, SCORPION_CLAW_EXECUTIONER, IMMORTAL_NECTAR_TOXIN_SAGE.
- **Contextual & Systemic Breakthroughs**:
    1. **Turn Pill Toxicity (`n.cult.toxic`) Into Symbiotic Gu Food (เลี้ยงหนอนกู่ด้วยพิษโอสถในกาย)!**:
        - Lore states: **"เลี้ยงหนอนกู่บรรพกาลไว้ในกาย... เปลี่ยนพิษในร่างเป็นพลังชีวิต · หากหนอนกู่ขาดอาหาร พิษกู่จะย้อนกัดกินตันเถียน"**.
        - Currently, VENOM_GU suffers pill `toxic` slowdown or ignores it without any feeding loop!
        - **Design Gem**: Every hour, a VENOM_GU cultivator's Symbiotic Gu digests `1.5` points of `n.cult.toxic` (or 1 low-tier toxic herb/venom sac if `toxic === 0` and stock > reserve) to charge **`+6 SYMBIOTIC_GU`** **and `+15% CULT.rate`**! Conversely, if both `n.cult.toxic === 0` and `SYMBIOTIC_GU < 10` for > 3 days, the starving Gu inflicts mild meridian gnawing (`-5% CULT.rate`, prompting the NPC to brew/consume toxic elixirs or hunt venomous beasts!).
    2. **`IMMORTAL_NECTAR_TOXIN_SAGE` (สายพิษสลับอมฤตช่วยชีวิต)**:
        - Wire its heal/support casts in `COMBAT` and `A.heal` to **invert active `poison`/`venomSeed` stacks on allies into instant healing bursts (`+6% MaxHP` per cleansed toxin stack)**!
    3. **`CORROSIVE_MIASMA_WEAVER` (สายทอหมอกพิษกัดกร่อนเกราะ)**:
        - Bypasses construct/golem biological poison resistance by corroding physical metal/stone (`applies armorbreak + reduces construct integrity by -8`)!

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6. [PATH-PUPPET-SHIFTER-06] Paths #26 & #18: SOUL_PUPPETRY (6 Subpaths) & PRIMORDIAL_SHIFTER (6 Subpaths)

6.1 SOUL_PUPPETRY (วิถีเชิดเงาวิญญาณและหุ่นเชิดด้ายชะตา — 6 Subpaths)

- **Subpaths**: PHANTOM_SILK_PUPPETEER, SOUL_STITCH_NEEDLE_MASTER, PAPER_EFFIGY_CONJURER, CORPSE_FLAME_MARIONETTE, MIRROR_SHADOW_TWIN, FATE_STRING_CONDUCTOR.
- **Critical Live Bug (MARIONETTE_SOUL_SWAP Self-Tether in js/combat.js)**:
  - MARIONETTE_SOUL_SWAP tethers target t to **linkedToId: t.id (itself!)** instead of selecting a second living enemy (C.enemies(W, f, a.id).find(e => e.id !== t.id)), reducing enemy-to-enemy spirit-silk damage sharing to **0%**! Fix target resolution and route through CULT.GRAND_DAO_MECHANICS.tetherMarionette.
  - Add **Severed-Silk Mental Recoil** (-8% Qi & brief exposed when an effigy/clone is destroyed by LIGHT_YANG / blade severing) to honor its supremeRisk!

6.2 PRIMORDIAL_SHIFTER (วิถีแปรรูปบรรพกาลและแปลงกาย — 6 Subpaths)

- **Subpaths**: AZURE_DRAGON_SCALE_SHIFTER, WHITE_TIGER_FANG_RAVAGER, VERMILION_PHOENIX_WING_ASCENDANT, BLACK_TORTOISE_SHELL_COLOSSUS, FIEND_APE_TITAN_MORPH, NINE_TAIL_MIRAGE_SHIFTER.
- **Homogenized Morph Buff & Missing Feral Sanity Threshold**:
  1. **Differentiate Subpath Morph Manifestations** (instead of giving every shifter identical atavismFrenzy +26% atk):
     - WHITE_TIGER_FANG_RAVAGER & FIEND_APE_TITAN_MORPH $\to$ atavismFrenzy (+atk / +crit / +poise break).
     - BLACK_TORTOISE_SHELL_COLOSSUS $\to$ Abyssal Tortoise Carapace (+30% def/mdef, +20% damage reflection, frontline FRONT_ROW anchor).
     - VERMILION_PHOENIX_WING_ASCENDANT $\to$ Nirvana Flame Plumage (+matk / +atk, burn aura, once-per-fight 28% HP Rebirth when BLOODLINE_ATAVISM >= 70).
     - NINE_TAIL_MIRAGE_SHIFTER $\to$ Nine-Tail Bewitching Mirage (+24% matk, +22% eva, charms/confuses attackers!).
     - AZURE_DRAGON_SCALE_SHIFTER $\to$ Dragon Scale Sovereign (balanced +18% atk/matk/def + LIGHTNING/WATER_ICE breath).
  2. **Feral Instinct Overrun (**สติมนุษย์ถูกสัญชาตญาณสัตว์ป่ากลืนกิน**)**: If BLOODLINE_ATAVISM >= 95 for 3+ consecutive rounds and composure <

50, enter **Feral Rampage** (+30% attack speed/damage, -15% defense, forces melee chase onto nearest enemy without retreating!).

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7. Implementation & Verification Checklist for GPT / Gemini (01F)

- [ ] **Fix `COMBAT.strike` Skill Scaling (`js/combat.js`)**: Replace broken ARRAY (1.00×) scaling on MERCANTILE_DAO and LORE_ARCHIVIST with ABACUS/trading and SCRIPTURE/BRUSH.
- [ ] **Fix `SOUL_PUPPETRY` Self-Tether (`js/combat.js`)**: Tether `t` to a distinct living enemy in MARIONETTE_SOUL_SWAP and unify with `tetherMarionette`.
- [ ] **Wire World & Sect Autonomous Loops**:
  - MERCANTILE_DAO: Real peer/caravan escrow contracts (`settleContract` wired!), EXT appraisal bonus, and trade `CULT.gain`.
  - LORE_ARCHIVIST & INK_LITERATI: Mid-combat LORE_INDEX observation ramp (TECHNIQUE_MIRROR_SCHOLAR), `annotateHeritageSutra` duty, and Righteous `integrity` scaling on LITERATI_INK.
  - WANDERING_DAO: 25-day indoor stagnation decay + high expedition volunteer drive + post-expedition Journey Epiphany `CULT.gain`.
  - SONIC_HARMONY: 3-Beat Harmonic Crescendo + `harmonic_pavilion` station routing + interrupt string-snap recoil.
  - VENOM_GU: Digest personal `cult.toxic` into SYMBIOTIC_GU & `CULT.rate` + IMMORTAL_NECTAR_TOXIN_SAGE poison-to-heal inversion + CORROSIVE_MIASMA_WEAVER construct corrosion.
  - PRIMORDIAL_SHIFTER: Subpath-specific Four-Symbol / Ape / Nine-Tail morph buffs + Feral Rampage at 95+ BLOODLINE_ATAVISM.